

Experiment 10

1. Experiment Title

Load and Display JSON Data Using `fetch()` and jQuery `$.getJSON()`

2. Software/Tools Required

- Visual Studio Code
- Google Chrome / Microsoft Edge
- HTML5
- CSS3
- JavaScript
- jQuery
- Live Server Extension

5. Theory

5.1 JSON

JSON (JavaScript Object Notation) is a lightweight format used for storing and exchanging structured data. It represents data using objects and arrays.

JSON is commonly used when exchanging data between a web application and a server.

5.2 `fetch()`

The JavaScript `fetch()` function is used to request data from a resource such as a JSON file or API.

The `response.json()` method converts the JSON response into JavaScript data that can be processed by the program.

5.3 jQuery `$.getJSON()`

jQuery provides the `$.getJSON()` method for loading JSON data.

It provides a simple way to request and process JSON data.

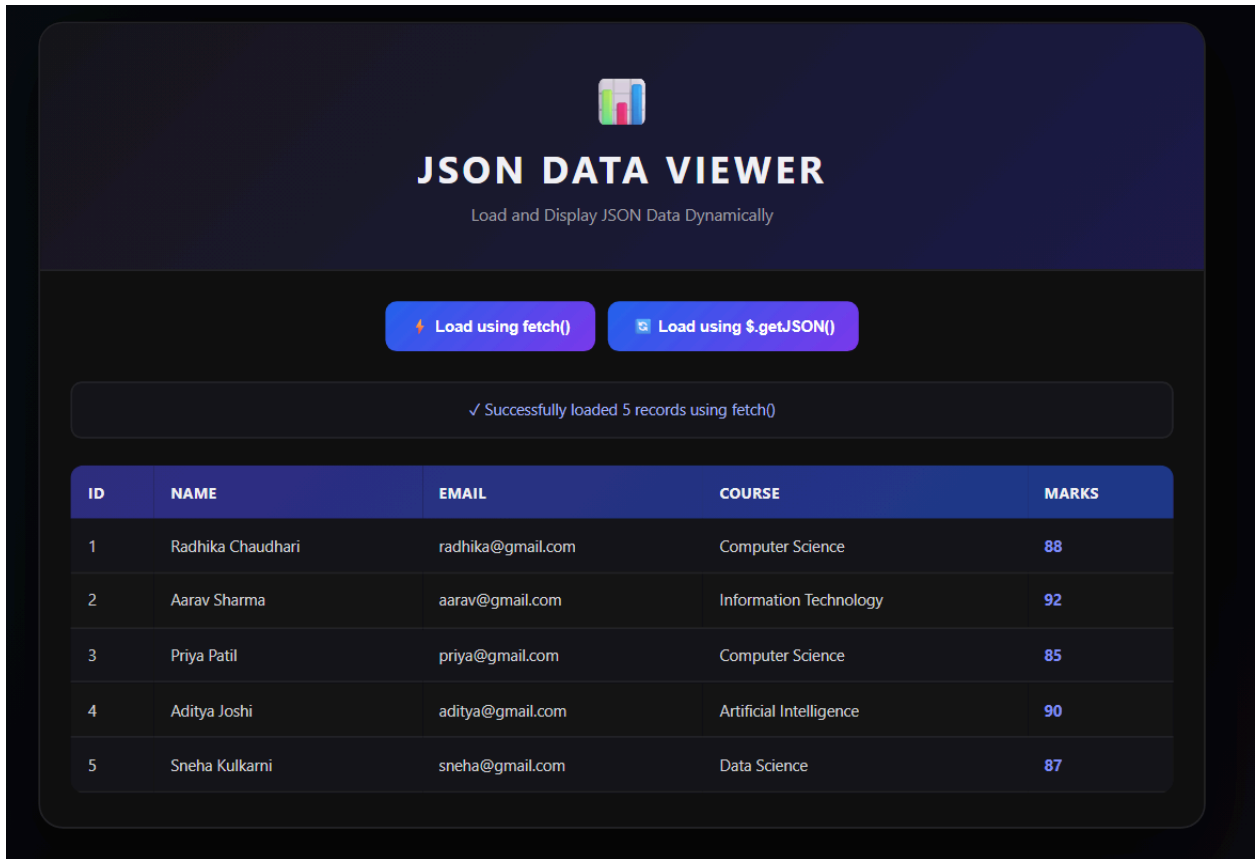
5.4 Dynamic Table Rendering

After loading the JSON data, JavaScript uses `forEach()` to process each object and creates table rows dynamically.

```
data.forEach(student => {  
  
    const row = document.createElement("tr");  
  
    row.innerHTML = `  
        <td>${student.id}</td>  
        <td>${student.name}</td>  
        <td>${student.email}</td>  
    `;  
  
    tableBody.appendChild(row);  
  
});
```

Thus, the table is generated automatically from the JSON data rather than entering every row manually.

4. Output



The screenshot shows a web application titled "JSON DATA VIEWER" with the subtitle "Load and Display JSON Data Dynamically". It features two buttons: "Load using fetch()" and "Load using \$.getJSON()". A status message indicates "✓ Successfully loaded 5 records using fetch()". Below this is a table with 5 rows of student data.

ID	NAME	EMAIL	COURSE	MARKS
1	Radhika Chaudhari	radhika@gmail.com	Computer Science	88
2	Aarav Sharma	aarav@gmail.com	Information Technology	92
3	Priya Patil	priya@gmail.com	Computer Science	85
4	Aditya Joshi	aditya@gmail.com	Artificial Intelligence	90
5	Sneha Kulkarni	sneha@gmail.com	Data Science	87

5. Result / Conclusion

The experiment was successfully implemented using HTML, CSS, JavaScript, JSON, and jQuery. JSON data was loaded using both the `fetch()` API and jQuery `$.getJSON()`, and the retrieved records were dynamically rendered into an HTML table. This experiment demonstrated how web applications can retrieve, process, and display structured JSON data dynamically.

Thus, the experiment successfully demonstrates loading and displaying JSON data using `fetch()` and `$.getJSON()`.